

Problem A: Let μ, λ be two probabilities on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Consider the functions $f(x, y) = x$, $g(x, y) = y$ defined from $(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \nu \equiv \mu \times \lambda)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. It is easy to see that these are measurable transformations, i.e random variables (Prove it if you don't see it!). Assume $f, g \in L^2(\mathbb{R}^2, \mathcal{B}(\mathbb{R}^2), \nu)$. Then show the following two facts:

$$\nu(\{f \in A\} \cap \{g \in B\}) \equiv \nu(\{(f, g) \in A \times B\}) = \nu(\{f \in A\})\nu(\{g \in B\}), \forall A, B \in \mathcal{B}(\mathbb{R})$$

$$\int f g \, d\nu = \left(\int f \, d\nu \right) \left(\int g \, d\nu \right).$$

[Recall our notation $\{f \in A\} \equiv \{(x, y) : f(x, y) \in A\} \equiv f^{-1}(A)$.

Hint: May have to use Cauchy Schwarz first!

Measure theory Trivia: Can you interpret this fact in terms of independent random variables?